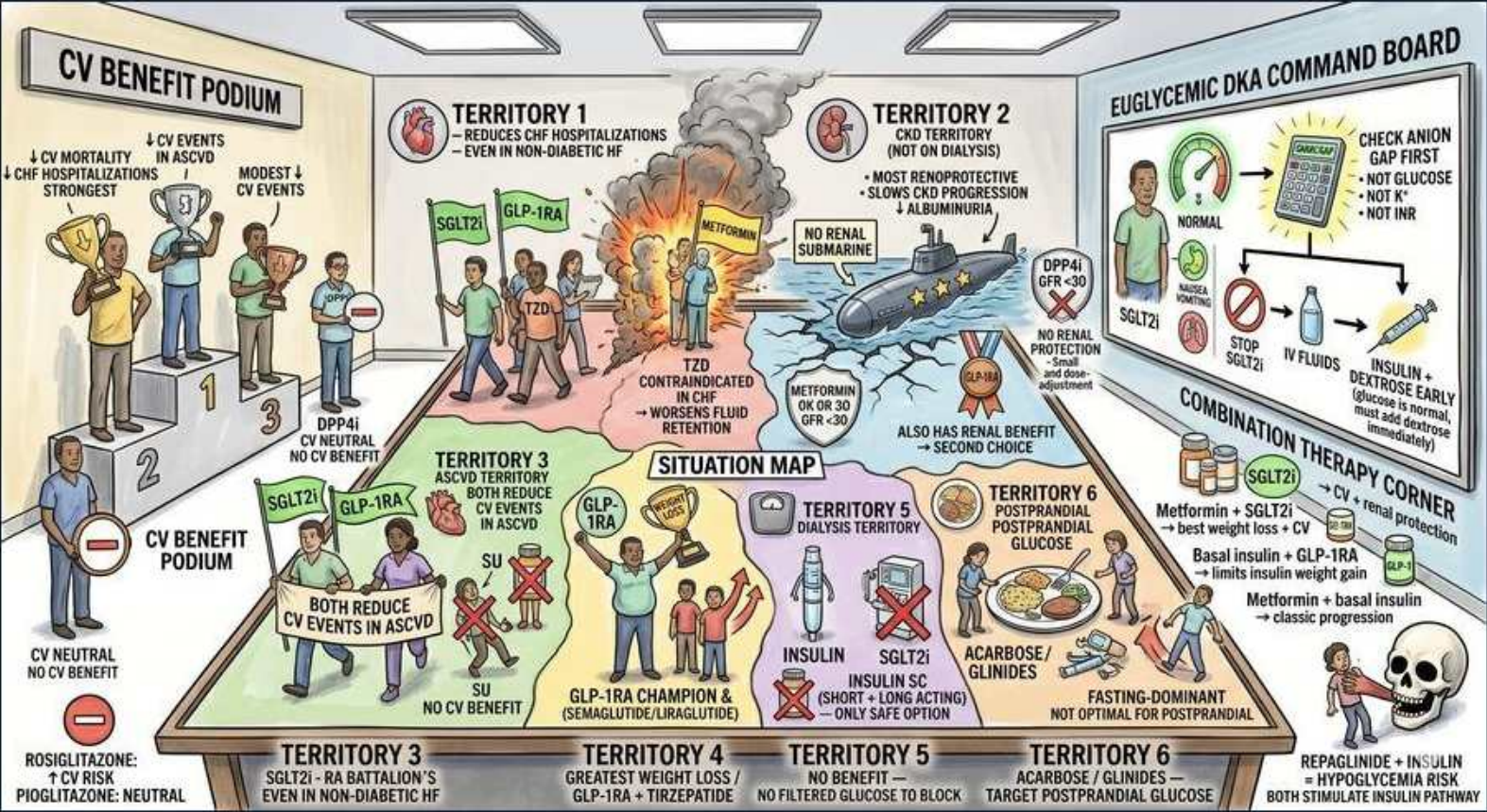


Diabetes — Scene B: The Diabetes War Room (CV Benefit & Comorbidity Drug Picks)



1. CV Benefit Podium

WHAT YOU SEE

Gold/silver/bronze medal podium (top-left) ranking CV/HF benefit: SGLT2i & GLP-1RA strongest, DPP4i CV-neutral.

WHAT IT ENCODES

Two classes have proven cardiovascular outcome benefit in T2DM: SGLT2 inhibitors and GLP-1 receptor agonists. SGLT2i are strongest for heart-failure hospitalization and renal/CV-death outcomes (EMPA-REG, DAPA-HF, EMPEROR, CREDENCE); GLP-1 RAs are strongest for atherosclerotic MACE (LEADER, SUSTAIN-6, REWIND). DPP-4 inhibitors and sulfonylureas are CV-NEUTRAL. Mnemonic: 'flozins win the heart-failure/kidney medal; glutides win the artery medal.'

BOARD TRAP

Wrong answer: 'add a DPP-4 inhibitor to reduce cardiovascular risk.' Discriminator — DPP-4i are CV-neutral and saxagliptin actually INCREASED heart-failure hospitalization (SAVOR-TIMI); for CV benefit choose SGLT2i or GLP-1 RA, not a gliptin.

2. Territory 1 — HFrEF

WHAT YOU SEE

Territory 1 (top center): heart with flames — SGLT2i + GLP-1RA reduce HF hospitalization, even in non-diabetic HF.

WHAT IT ENCODES

In heart failure (reduced AND preserved EF), SGLT2 inhibitors reduce HF hospitalization and CV death even in patients WITHOUT diabetes — they are now a guideline pillar of HFrEF therapy (with ARNI, beta-blocker, MRA). Dapagliflozin (DAPA-HF) and empagliflozin (EMPEROR-Reduced) are first choice. Mnemonic: 'a flozin for every failing pump.'

BOARD TRAP

Wrong answer: 'choose a GLP-1 RA for a diabetic with HFrEF because it lowers glucose and weight.' Discriminator — GLP-1 RAs do NOT share the robust HF-hospitalization benefit of SGLT2i; the dedicated HF agent is the SGLT2 inhibitor. Pick the flozin for the failing heart.

3. TZD contraindicated in HF

WHAT YOU SEE

Explosion just right of Territory 1: TZD CONTRAINDICATED in CHF — worsens fluid retention.

WHAT IT ENCODES

Thiazolidinediones activate PPAR-γ → renal sodium/water retention → edema, weight gain, and precipitation/worsening of heart failure; they are contraindicated in symptomatic (NYHA III–IV) HF. The effect is worse when combined with insulin. Mnemonic: 'PPAR-γ pours fluid into the failing heart.'

BOARD TRAP

Wrong answer: 'pioglitazone is acceptable in mild heart failure since it's CV-neutral on MACE.' Discriminator — regardless of MACE neutrality, TZD fluid retention worsens HF; both pio- and rosiglitazone are contraindicated in symptomatic HF. Use SGLT2i instead.

4. Territory 2 — CKD (not dialysis)

WHAT YOU SEE

Territory 2 (top center-right): submarine — CKD not on dialysis, SGLT2i most renoprotective.

WHAT IT ENCODES

In diabetic CKD (not yet on dialysis), SGLT2 inhibitors are most renoprotective: they reduce intraglomerular pressure (afferent arteriolar constriction), slow GFR decline and progression to ESRD, and lower albuminuria (CREDENCE, DAPA-CKD). They can be initiated to low eGFR (~20) for renal protection even though glucose-lowering wanes. Add a GLP-1 RA for additional CV/glycemic benefit. Mnemonic: 'flozin = the kidney's bodyguard.'

BOARD TRAP

Wrong answer: 'avoid SGLT2i in a patient with eGFR 35 and albuminuria because they don't lower glucose much at that level.' Discriminator — the renoprotective indication persists at low eGFR even when the glycemic effect fades; you keep/start the SGLT2i for kidney protection. Don't withhold it just because A1c drop is small.

5. Metformin in CKD

WHAT YOU SEE

Metformin pill block below the submarine: OK if eGFR>30, second choice / avoid in low eGFR.

WHAT IT ENCODES

Metformin remains acceptable in CKD if eGFR >30 (reduce dose / use caution at 30–45); contraindicated below 30 because impaired clearance raises lactic acidosis risk. Always hold around iodinated contrast and acute illness/AKI. Mnemonic: 'metformin lives above an eGFR of 30.'

BOARD TRAP

Wrong answer: 'stop metformin once eGFR drops below 45 or 60.' Discriminator — the hard cutoff is eGFR 30, not 45/60 (dose-reduce, don't stop, at 30–45). Choosing the higher numbers is the classic numeric trap; also remember to hold for contrast.

6. Territory 5 — Dialysis

WHAT YOU SEE

Territory 5 (center-right band): dialysis territory — insulin SC only safe option, SGLT2i red X.

WHAT IT ENCODES

In ESRD on dialysis, INSULIN is the safe and preferred glucose-lowering agent (titrate carefully — insulin clearance falls, hypoglycemia risk rises, and needs often drop). Most oral agents are contraindicated/cleared poorly, and SGLT2i are not indicated/effective on dialysis. Mnemonic: 'on dialysis, the answer is insulin.'

BOARD TRAP

Wrong answer: 'continue the SGLT2 inhibitor for renal protection after the patient starts hemodialysis.' Discriminator — SGLT2i renal benefit applies to non-dialysis CKD; on dialysis they are not indicated. Switch to insulin, the safe option in ESRD.

7. Territory 3 — ASCVD

WHAT YOU SEE

Territory 3 (mid-left band): ASCVD — both SGLT2i & GLP-1RA reduce CV events; SU red X (no CV benefit).

WHAT IT ENCODES

In established atherosclerotic CV disease (prior MI, stroke, PAD), BOTH GLP-1 RAs and SGLT2i reduce major adverse CV events; GLP-1 RAs have the strongest evidence for atherosclerotic events while SGLT2i excel at HF/renal endpoints. Either is an appropriate, guideline-endorsed add-on regardless of A1c. Mnemonic: 'arteries clogged → glutide (or flozin), never a sulfonylurea.'

BOARD TRAP

Wrong answer: 'a sulfonylurea is a reasonable second agent in a patient with ASCVD because it lowers A1c well.' Discriminator — SUs have NO CV benefit (and cause hypoglycemia/weight gain); in ASCVD select a GLP-1 RA or SGLT2i with proven MACE reduction.

8. Territory 4 — Weight loss champion

WHAT YOU SEE

Territory 4 (center band): GLP-1RA weight-loss champion trophy (semaglutide/liraglutide, tirzepatide).

WHAT IT ENCODES

Greatest weight loss comes from incretin-based injectables: GLP-1 RAs (semaglutide ~10–15%, liraglutide) and the dual GIP/GLP-1 agonist tirzepatide (up to ~20%, the most potent). SGLT2i add modest loss (~2–3 kg). These are first-line when obesity dominates the phenotype. Mnemonic: 'tirzepatide = two incretins = top of the weight-loss podium.'

BOARD TRAP

Wrong answer: 'add basal insulin, a sulfonylurea, or a TZD when weight loss is the priority.' Discriminator — insulin, SU, glinides, and TZD all cause weight GAIN, the opposite of the goal. Choose a GLP-1 RA or tirzepatide for the weight-loss phenotype.

9. Territory 6 — postprandial glucose

WHAT YOU SEE

Territory 6 (right band): postprandial glucose — acarbose / glinides.

WHAT IT ENCODES

When the problem is POSTPRANDIAL hyperglycemia (high after-meal spikes with acceptable fasting glucose), the targeted agents are acarbose (alpha-glucosidase inhibitor — delays carb absorption) and glinides/meglitinides (short-acting secretagogues dosed before each meal). Rapid-acting prandial insulin also targets this. Mnemonic: 'meal-time spikes → meal-time drugs (acarbose, glinides).'

BOARD TRAP

Wrong answer: 'use acarbose or a glinide to fix isolated FASTING hyperglycemia.' Discriminator — these target POSTPRANDIAL, not fasting, glucose; fasting hyperglycemia is better addressed by metformin (hepatic output) or basal insulin. Also acarbose's GI flatulence/diarrhea limits adherence.

10. Euglycemic DKA board

WHAT YOU SEE

'Euglycemic DKA Command Board' (top-right): check anion gap first, stop SGLT2i, IV fluids, insulin + dextrose early.

WHAT IT ENCODES

SGLT2 inhibitors can precipitate EUGLYCEMIC DKA (ketoacidosis with glucose often <200–250) via glucosuria-driven low insulin and increased glucagon/ketogenesis, typically triggered by surgery, fasting, illness, or alcohol. Diagnose by the ANION GAP and ketones, NOT glucose. Treatment: IV fluids, insulin infusion WITH dextrose (since glucose is already low/normal), and stop the SGLT2i; hold SGLT2i ~3 days before elective surgery. Mnemonic: 'normal sugar, sky-high gap — think flozin DKA.'

BOARD TRAP

Wrong answer: 'a normal blood glucose rules out DKA in a patient on an SGLT2 inhibitor with vomiting and an elevated anion gap.' Discriminator — euglycemic DKA presents with normal/mildly elevated glucose; you must check the anion gap and serum/urine ketones. Don't be falsely reassured by a normal glucose.

11. Combination therapy corner

WHAT YOU SEE

Combination Therapy Corner (right side): metformin + SGLT2i, basal insulin + GLP-1RA, metformin + basal insulin.

WHAT IT ENCODES

Rational combinations keep metformin as the backbone and add a mechanistically COMPLEMENTARY agent: metformin + SGLT2i (good weight loss, CV/renal benefit), metformin + GLP-1 RA (max weight loss/CV benefit, limits insulin-related gain), metformin + basal insulin (when more glucose lowering is needed). Mnemonic: 'build on metformin with a different mechanism.'

BOARD TRAP

Wrong answer: 'combine two agents from the same pathway (e.g., DPP-4i + GLP-1 RA, or SU + glinide) to intensify therapy.' Discriminator — same-pathway combos are redundant (incretin + incretin) or dangerously additive (two secretagogues → hypoglycemia). Stack DIFFERENT mechanisms instead.

12. No SU + glinide combo / repaglinide+insulin

WHAT YOU SEE

Skull (bottom-right): repaglinide + insulin and SU + glinide both stimulate insulin secretion — hypoglycemia risk.

WHAT IT ENCODES

Never combine two insulin secretagogues (sulfonylurea + glinide) or a secretagogue plus prandial insulin (repaglinide + mealtime insulin) — they share the K-ATP-closing mechanism, so the effect is redundant and additively raises hypoglycemia risk with no extra efficacy. Mnemonic: 'two insulin-squeezers = double the hypo, none of the benefit.'

BOARD TRAP

Wrong answer: 'add a glinide to a patient already on a sulfonylurea for better postprandial control.' Discriminator — both are secretagogues acting via the same channel; combining them just compounds hypoglycemia/weight gain. Add a different-mechanism agent (metformin, SGLT2i, GLP-1 RA) instead.

13. Rosiglitazone vs pioglitazone CV

WHAT YOU SEE

Bottom-left near podium: rosiglitazone increased CV risk / pioglitazone CV-neutral.

WHAT IT ENCODES

Among TZDs, rosiglitazone raised concern for increased myocardial infarction/CV risk (historic meta-analyses), whereas pioglitazone is CV-NEUTRAL (and may modestly reduce stroke/MI in some data) — making pioglitazone the preferred TZD. BOTH still cause fluid retention/edema, weight gain, fractures, and HF risk; pioglitazone additionally carries a bladder-cancer signal. Mnemonic: 'PIO is the okay one; ROSI raised the red flag.'

BOARD TRAP

Wrong answer: 'rosiglitazone is the safer thiazolidinedione for cardiovascular risk.' Discriminator — it's the reverse: pioglitazone is the CV-safer TZD; rosiglitazone carried the increased CV/MI concern. Either way, avoid both in heart failure due to fluid retention.